

Math 2

Date / / Subject

* Separation of Variables:-

Solve the D.E:-

Ex) $x \frac{dy}{dx} = y$

$$\frac{1}{y} dx = \frac{1}{x} dx$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\ln y = \ln x + C$$

$$\ln y = \ln x + \ln C$$

للتبسيط، نغضو عن الثابت C بـ $\ln C$

من خواص اللوغاريتمات الجمع يتحول لضرب

$$\ln y = \ln(C \cdot x)$$

$$y = Cx$$

Find general solve for:-

$$x(4-x) \frac{dy}{dx} - y = 0$$

$$x(4-x) \frac{dy}{dx} = y$$

Separation Variable & Integration

$$\int \frac{dy}{y} = \int \frac{dx}{x(4-x)}$$

* الحروف الأيمن يحتاج Partial fractions

$$\frac{1}{x(4-x)} = \frac{A}{x} + \frac{B}{4-x}$$

$$1 = A(4-x) + Bx$$

$$x=0 \rightarrow 1 = 4A \quad \boxed{A = \frac{1}{4}}$$

$$x=4 \rightarrow 1 = 4B \quad \boxed{B = \frac{1}{4}}$$

$$\int \frac{1}{y} dy = \int \frac{1}{4x} + \frac{1}{4(4-x)} dx$$

$$= \frac{1}{4} \int \frac{1}{x} + \frac{1}{4} \int \frac{1}{4-x} dx$$

$$\ln y = \frac{1}{4} \ln x + \frac{1}{4} - \ln(4-x) + \ln C$$

$$\ln y = \ln x^{\frac{1}{4}} - \ln(4-x)^{\frac{1}{4}} + \ln C$$

note
 $\ln a - \ln b = \ln \frac{a}{b}$

$$y = C \left(\frac{x}{4-x} \right)^{\frac{1}{4}}$$

Ex) $\frac{dy}{dx} = x^5 + 3e^{-2x}$

Separation Variable & integration

$$\int dy = \int x^5 + 3e^{-2x} dx$$

$$y = \frac{x^6}{6} + 3 \int e^{-2x} dx$$

$$y = \frac{x^6}{6} + \frac{-3}{2} \int -2e^{-2x} dx$$

$$y = \frac{x^6}{6} - \frac{3}{2} e^{-2x} + C$$

note
 $\int a^x e^{bx} dx = \frac{a^x e^{bx}}{b} + C$
 $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Ex) $x(y-3) \frac{dy}{dx} = 4y$

Separation Variable & Integration

$$\int \frac{y-3}{y} dy = 4 \int \frac{1}{x} dx$$

$$\int (1 - \frac{3}{y}) dy = 4 \int \frac{1}{x} dx$$

$$y - 3 \ln y = 4 \ln x + \ln c$$

$$y - 3 \ln y = \ln x^4 + \ln c$$

$$y = \ln x^4 + \ln c + \ln y^3 \rightarrow y = \ln(c \cdot x^4 \cdot y^3)$$

* Homogenous Equation:-

* تكون المعادلة متجانسة إذا كانت على الصورة $y = f(\frac{y}{x})$ وكل الحدود لها نفس الدرجة

* طريقة الحل هي استخدام التعويض $y = xV$ وبعد التعويض نستخرج فضل المتغيرات

Ex) $x \frac{dy}{dx} = x + 2y$ Subject to condition $y=6$, When $x=1$

The Equation is homogenous $\rightarrow y = xV$ (نفاصل الطرفين (حاصل ضرب اثنين)

$$\frac{dy}{dx} = V + x \frac{dV}{dx}$$

$$x(V + x \frac{dV}{dx}) = x + 2xV$$

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$$xV + x^2 \frac{dV}{dx} = x + 2xV$$

$$x^2 \frac{dV}{dx} = x + 2xV - xV$$

$$x^2 \frac{dV}{dx} = x + xV \quad (\div x)$$

$$x \frac{dV}{dx} = 1 + V$$

Separation and Integration

$$\int \frac{dV}{1+V} = \int \frac{dx}{x}$$

$$\ln(1+V) = \ln x + \ln c$$

$$\ln(1+V) = \ln(cx)$$

$$1+V = cx \rightarrow V = cx - 1$$

نضع الـ V لأصلها $\frac{y}{x}$

$$\frac{y}{x} = cx - 1 \rightarrow y = x(cx - 1)$$

$$6 = c - 1 \rightarrow c = 7$$

$$y = x(7x - 1)$$

Ex) $2xy \frac{dy}{dx} + (x^2 + y^2) = 0$

The Equation is homogenous

Let $y = xV$ $\rightarrow$ نشتق الطرفين

$$\frac{dy}{dx} = V + x \frac{dV}{dx}$$

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$$2X(XV)(V + X \frac{dV}{dX}) + (X^2 + X^2 V^2) = 0$$

$$2X^2 V^2 + 2X^3 V \frac{dV}{dX} + X^2 + X^2 V^2 = 0$$

$$3X^2 V^2 + X^2 = -2X^3 V \frac{dV}{dX} \quad (+X^2)$$

$$3V^2 + 1 = -2X V \frac{dV}{dX}$$

Separation and Integration

$$\int \frac{-2V}{1+3V^2} dV = -\int \frac{1}{X} dX$$

$$\frac{1}{3} \int \frac{6V}{1+3V^2} dV = -\ln X + \ln C$$

$$\frac{1}{3} \ln(1+3V^2) = \ln\left(\frac{C}{X}\right)$$

$$\ln(1+3V^2)^{\frac{1}{3}} = \ln\left(\frac{C}{X}\right)$$

$$(1+3V^2)^{\frac{1}{3}} = \frac{C}{X}$$

* بتكعب الطرفين وارباع V لاصلاها $\frac{y}{x}$

$$1+3V^2 = \left(\frac{C}{X}\right)^3$$

$$3\left(\frac{y}{x}\right)^2 = \left(\frac{C}{x}\right)^3 - 1$$

$$y^2 = \frac{x^2}{3} \left(\left(\frac{C}{x}\right)^3 - 1 \right)$$

* Linear D.E :-

$$\frac{dy}{dx} + P(x)y = Q(x)$$

* طريقة الحل

① نكتب قيم الـ $P(x)$ و $Q(x)$

$$I.f = e^{\int P(x) dx}$$

② نحسب معامل التكامل (Integrating factor)

$$(I.f)y = \int (I.f)Q(x) dX + C$$

③ نوجد الحل عن طريق المعادلة:

(EX) $\frac{dy}{dx} + \frac{1}{x}y = x^2$

The Equation is linear

$$P(x) = \frac{1}{x}, \quad Q(x) = x^2$$

$$I.f = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$x \cdot y = \int x \cdot (x^2) dX + C \rightarrow xy = \int x^3 dX + C$$

$$xy = \frac{x^4}{4} + C \rightarrow y = \frac{1}{4}x^3 + \frac{C}{x}$$

(EX) $\frac{dy}{dx} + 2xy = 2xe^{-x^2}$

The Equation is linear

$$P(x) = 2x, \quad Q(x) = 2xe^{-x^2}$$

$$I.f = e^{\int 2x dx} = e^{x^2}$$

$$e^{x^2} \cdot y = \int e^{x^2} \cdot (2x e^{-x^2}) dx + C$$

$$e^{x^2} \cdot e^{-x^2} = e^0 = 1$$

في الضرب نجمع الأسس للدالة الأسية

$$e^{x^2} \cdot y = \int 2x dx + C$$

$$e^{x^2} \cdot y = x^2 + C \quad \Rightarrow y = \frac{x^2 + C}{e^{x^2}}$$

(Ex) $3 \frac{dy}{dx} + 2y = 0$

, $(y=10)$ when $(x=0)$

بالضرب على 3 لنصل إلى شكل المعادلة الخطية

$$\frac{dy}{dx} + \frac{2}{3}y = 0$$

The Equation is Linear

$$P(x) = \frac{2}{3}, \quad Q(x) = 0$$

$$I.F = e^{\int \frac{2}{3} dx} = e^{\frac{2}{3}x}$$

$$e^{\frac{2}{3}x} \cdot y = \int e^{\frac{2}{3}x} \cdot (0) dx + C$$

$$e^{\frac{2}{3}x} \cdot y = C$$

$$y = \frac{C}{e^{\frac{2}{3}x}} = C e^{-\frac{2}{3}x}$$

$$(x=0), (y=10)$$

$$10 = C e^0 \quad \Rightarrow 10 = C \quad \Rightarrow C=10$$

$$y = 10 e^{-\frac{2}{3}x}$$